Introductory Econometrics with Recitation — Quiz 5

May 19, 2026

Name: ________________ **ID:** ________________

1. (**17 points**) The following table reports two panel regressions for 50 cities from 2017 to 2023. Each observation is a city-year, and all variables in this table are measured at the city-year level. The variables are defined as follows:

Variable Name	Description
InjuryRate	Number of e-scooter injuries per 100,000 residents in a city-year.
HelmetFine	Fine amount, in dollars, for riding an e-scooter without a helmet in that city-year.
Age18	= 1 if the city's minimum riding age is 18 in that year, and 0 otherwise.

Table 1: Impact of Helmet Fines on E-Scooter Injuries

	Dependent variable: *InjuryRate*	
Regressor	(1)	(2)
ln(HelmetFine)	-1.80	-1.60
	(0.55)	(0.50)
Age18	-0.52	-0.46
	(0.22)	(0.21)
City Fixed Effects	Yes	Yes
Year Fixed Effects	No	Yes
Clustered Standard Errors	Yes	Yes
Observations	350	350
R^2	0.68	0.72

Notes: Standard errors are reported in parentheses.

(a) (**8 points**) Using Model (1), test whether the coefficient on ln(HelmetFine) is statistically significant at the 5% level. Interpret the coefficient in words.

(b) (**3 points**) Why might the regression include city fixed effects? Give one example of an omitted variable that can be captured by city fixed effects.

(c) (**3 points**) Why might Model (2) include year fixed effects? Give one example of an omitted variable that can be captured by year fixed effects.

(d) (**3 points**) How should standard errors be clustered in this regression? Explain briefly.

2. (**13 points**) A researcher uses data on 700 private-sector employees to study whether employees are allowed to work remotely at least two days per week. The variables are defined as follows:

Variable Name	Description
$Remote_i$	= 1 if employee i is allowed to work remotely at least two days per week, and 0 otherwise.
$Experience_i$	Years of work experience for employee i.
$Female_i$	= 1 if employee i is female, and 0 otherwise.

Table 2: Regression Models for Remote Work Eligibility

	Dependent variable: *Remote*		
Regressor	Probit	Logit	LPM
Experience	0.120	0.200	0.025
	(0.030)	(0.050)	(0.007)
Female	0.450	0.780	0.090
	(0.280)	(0.490)	(0.060)
Constant	-1.250	-2.100	0.280
	(0.270)	(0.470)	(0.060)
Observations	700	700	700

Notes: Standard errors are reported in parentheses.

(a) (**3 points**) Using the probit model, compute the predicted probability of $Remote = 1$ for Liam, a man with 10 years of experience.

(b) (**3 points**) Using the logit model, compute the predicted probability for Liam.

(c) (**3 points**) Using the LPM, compute the predicted probability for Liam.

(d) (**4 points**) Compare the linear probability model (LPM) with the probit model. Provide one advantage and one disadvantage of using the LPM.